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Draft Project Proposal

Phase 1: The Itch

- What is a task you have to do repeatedly that is boring, difficult, or time-consuming?
    - For this project, we decided to go down Pathway A, the more academic and scholarly pathway. We are focusing on a problem that occurs in scientific research laboratories.
    - In fish research laboratories, like Serena's lab, manually counting the number of brine shrimp in petri dish images that are used to feed the in-house fish is extremely time-consuming and tedious.
    - Researchers have to analyze many images for every round of the feeding protocol. This makes the manual counting process slow, repetitive, and prone to human counting errors.
- What is a "truth" in your field that you think AI might get wrong? (This is a great starting point for a stress-test project).
    - While detecting and counting the brine shrimp, the AI might not understand at first that not every small object or shape in the images are brine shrimp.
    - AI systems may incorrectly detect debris, bubbles, or shadows as shrimp, possible leading to inaccurate object detection and incorrect counts.

Phase 2: The Product

- If this project succeeds, what will you actually *have* at the end?
    - A working AI detection and counting pipeline for petri dish images.
    - The AI pipeline will output a report that displays the brine shrimp count for each image.
    - The AI pipeline will also display the petri dish images with the detected shrimp in bounding boxes, to use for manual checking.

- Who is the "User"? (Is it you? A fellow student? A professor? A client?)
    - Researchers working with brine shrimp and petri dish image data.

Phase 3: The "Centaur" Factor

- What part of this project requires **human** judgment?
    - Labeling training images correctly.
    - Checking whether the detected objects are actually brine shrimp.
- What part of this project requires **AI** speed or synthesis?
    - Processing large amounts of images quickly.
    - Automatically detecting and counting the brine shrimp.
- *Crucial Question:* If you took the AI away, would the project be impossible? If you took the Human away, would the project be dangerous/inaccurate? (We want "Yes" to both).
    - Yes, to both as the most accurate detection and counting system requires both AI detection and human verification.
    - Without AI, counting shrimp across many images would take a very long time and be inefficient.
    - Without human review, the AI could have incorrect counts for the images.

Phase 4: The Resources

- What "Knowledge" does the AI need? (e.g., "It needs to know the rules of baseball," "It needs to know the last 10 years of inflation data," "It needs to know the text of *Hamlet*").
    - The AI needs labeled training data that includes many images with bounding boxes around brine shrimp in petri dish images.
- Do you have access to this data?
    - Yes. We have access to petri dish images containing brine shrimp.
    - These images can be labeled and used to train the AI detection system.